

Closed-Loop BCI-Triggered System with Soft-Robotic Glove for after-stroke rehabilitation

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1 Clinical Need

Stroke represents a major cause of long-term neurological disability globally. As evidenced by epidemiological data from the World Stroke Organization [1], the socio-health impact of this pathology is constantly growing: over 93 million cases prevalent, some 160 million years of life lost to disability (DALYs) and a cost estimated global economic of more than US\$ 890 billion.

Between 55% and 75% of survivors have chronic upper limb motor deficit which compromises autonomy in daily activities (ADL) [12]. Donkor [2] contextualize this figure in the perspective of quality of life: stroke is the second cause of death at the world and causes severe permanent disability in about half of the survivors, with high impact on autonomy, working life and the care burden on family members.

This combination of high prevalence and high residual disability justifies the search for advanced technological solutions that can deliver portable therapy with an intensity comparable to that of departments of specialized rehabilitation. Recent clinical research indicates that the effectiveness of traditional rehabilitation treatments is severely limited by a therapeutic dosage deficit. The systematic analysis conducted by Newton et al. [3] highlights how the actual time spent on active therapeutic exercise (time on task) within conventional sessions varies on average between 8 and 44 minutes per day — significantly lower than recommended doses to induce cortical reorganization stable. A clinical paradox emerges: patients with more severe impairment receive doses of significantly lower active therapy precisely because of the high effort required without tools modular assistance.

2 Current Standard of Care and Its Limitations

The current standard of post-stroke upper limb rehabilitation combines conventional physiotherapy with commercially available robotic orthoses. Both approaches share a critical failure mode: they address the peripheral component of motor recovery while neglecting its central requirement.

Newton et al. [3] quantify this dose gap: effective active therapy time in conventional care averages 8–44 minutes per day — far below the repetition volume required to induce stable cortical reorganization. After discharge, most patients receive virtually no structured rehabilitation.

The robotic hardware available today does not close this gap. Wang et al. [7] compared a passive soft-robotic glove against repetitive transcranial magnetic stimulation (rTMS) in patients with severe upper limb dysfunction. The passive glove improved Fugl-Meyer Assessment scores (FMA-UE) by facilitating joint mobilization, but produced no significant changes in Resting Motor Threshold (RMT) or Motor Evoked Potential (MEP) amplitude — the markers of cortical synaptic excitability. The glove moves the hand, it does not engage the brain.

The neurobiological explanation is well established. Long-term motor recovery depends on Hebbian plasticity (Spike-Timing-Dependent Plasticity, STDP): stable synaptic modifications require temporal coincidence — within approximately 300 ms — between the pre-synaptic signal (motor cortex activation associated with movement intention) and the post-synaptic afferent feedback generated by the actual limb movement [4, 5]. In the μ (8–12 Hz) and β (13–30 Hz) bands, Motor Imagery manifests as event-related desynchronization (ERD) over contralateral motor areas (C3/C4). When a BCI detects this ERD and uses it as an instantaneous trigger, the Hebbian coincidence window is

realized. A random or sham activation produces no significant increase in β -band corticomuscular coherence [5] and no relevant clinical improvement [6].

The biomarker of this reorganization is measurable in real time: Rustamov et al. [11] demonstrate that theta-gamma cross-frequency coupling (CFC) over C3/C4 increases proportionally to motor recovery across BCI sessions, confirming CFC as a neurobiological proxy for treatment efficacy. The mechanism exists, is validated, and is quantifiable. The gap is not biological — it is a delivery problem.

3 The Human Limitation Being Addressed

The mechanism described above requires three conditions that no human operator can reliably provide in an unsupervised home setting:

(1) Continuous presence. The Hebbian contingency must be enforced at every repetition across sessions of 30–45 minutes, 3–5 times per week, for months. A trained therapist cannot be physically present at the patient’s home at this frequency — this is precisely the dose gap documented by Newton et al. [3].

(2) Sub-300 ms reactive latency. The BCI trigger must fire within the Hebbian coincidence window [5]. To allow a consistent satisfaction of this condition, a robotic solution is the most reliable means of automatic response.

(3) Operator-independent consistency. Outcome variability in conventional rehabilitation is partially attributable to inter-therapist inconsistency in task delivery and feedback timing [3]. A closed-loop BCI system applies identical contingency criteria at every trial, eliminating this source of variance. Ren et al. [9] confirm in a meta-analysis of 10 RCTs (290 patients) that BCI-triggered intervention outperforms conventional therapy (SMD = 0.50, 95 % CI: 0.26–0.73), with effect size directly linked to contingency fidelity.

The robot is not a substitute for clinical judgment — the therapist remains in the loop asynchronously via remote dashboard. It is a substitute for physical presence and consistent and automatic reactivity.

4 Proposed Robotic Solution

To overcome the issue, the proposed solution is a BCI-Triggered system with a Soft Robotic Glove. This is the direct consequence of the need to satisfy the time gap consistency and the continuative dose during years. The device is classified as a wearable, domiciliary and supervised. It is used in autonomous session of 30-45 minutes per 3-5 times per week, without the clinician presence. An GUI is used to set up and start up the session, and the clinician monitors during time with a claud communication.

4.1 Kinematic Architecture

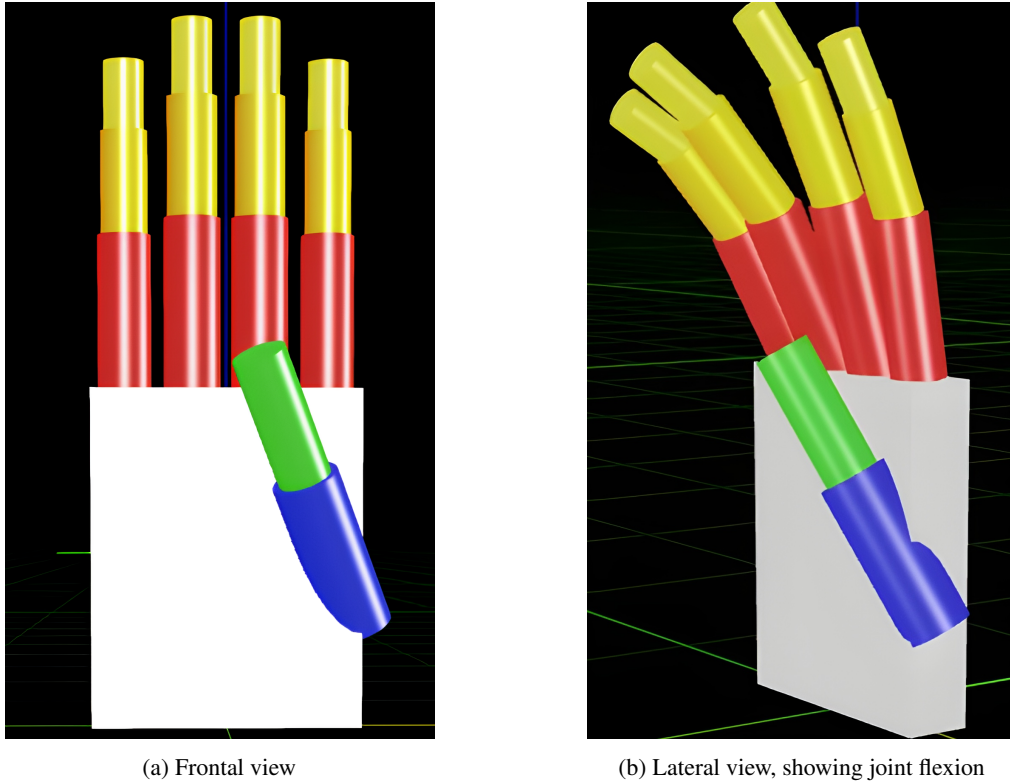
The architecture is a soft-robotic pneumatic wearable glove, modelled as a system of serial kinematic chain anchored to a fixed palm [12, 15]. The choice of using a pneumatic network (PneuNet) is dictated by the need of intrinsic security and passive adaptability to variable geometries in an unsupervised environment.

The geometric model replicates human anatomy with 3 Degrees of Freedom (DoFs) for each of the 4 fingers (MCP, PIP, and DIP joints) and 2 DoFs for the thumb (CMC and MCP joints), reaching a total of 14 kinematic DoFs. However, the system is intentionally under-actuated, employing a single independent pneumatic channel per digit to yield 5 active DoFs. The complexity-benefit trade-off of this design is governed by three primary factors:

1. **Biological Synergies:** Neurophysiological studies (e.g., Santello et al. [17]) demonstrate that the central nervous system controls hand grasping through low-dimensional synergistic patterns rather than isolated joint control. Actuating 14 joints independently is redundant for post-stroke

rehabilitation, which focuses on restoring gross functional synergies for Activities of Daily Living (ADLs).

2. **Embodied Intelligence:** The PneuNet architecture exploits morphological computation. Upon pressurization, fluid redistribution within the elastomeric chambers naturally conforms to the object's geometry, bypassing the need for high-density sensor arrays or complex multi-joint control algorithms.
3. **Usability and Weight Constraints:** Implementing 14 active lines would introduce excessive hardware overhead. Minimizing weight is crucial to prevent muscular fatigue and abnormal compensatory movements at the shoulder, ensuring autonomous, single-handed donning by hemiparetic patients at home.



Figur 1: URDF kinematic model of the robotic glove.

Mathematically, this under-actuation defines a constrained kinematic mapping where the joint space $q \in \mathbb{R}^{14}$ is coupled to the actuator pressure space $u \in \mathbb{R}^5$ via a non-linear vector function:

$$q = g(u) \tag{1}$$

where the passive elasticity of the silicone elastomer acts as a virtual restoring spring, guaranteeing a unique, statically stable equilibrium configuration for each input pressure vector u .

4.2 Sensing and Biomechanical Safety

System sensing is categorized into two levels: neurological sensing (EEG) and local biomechanical sensing. To ensure patient safety in the absence of a clinician, the firmware implements three quantitative mechanisms in compliance with active medical device standards (IEC 60601-1) and IEC 62304 software guidelines. The system is designed to achieve the Performance Level required by ISO 13849:

- **Admittance Control** ($F \rightarrow v$): Integrated palmar piezoelectric sensors measure interaction forces, dynamically modulating pneumatic pressure to ensure glove compliance; the system automatically yields to spastic resistance, prioritizing patient safety and joint protection. Kinematic joint boundaries defined in the URDF model serve as software-enforced virtual fixtures.
- **Fail-Safe EEG**: The system continuously monitors electrode impedance and signal quality (SNR). In the event of signal loss (1 s timeout), all pneumatic valves are exhausted within 200 ms, ensuring the device cannot execute non-intentional autonomous movements.
- **Emergency Stop**: A physical, high-visibility emergency button is located on the device controller. Accessible with the patient’s unaffected hand, it immediately cuts power to the valves and exhausts residual pneumatic pressure, independently of the software state.

While EEG-based BCI for post-stroke rehabilitation is a safe and clinically established modality, the primary challenge lies in achieving robust decoding rather than biological feasibility. Cortical lesions often trigger neuroplastic reorganization, which can alter the amplitude and spatial topography of Motor Imagery signals. To address this inter-patient variability, our system employs an adaptive Linear Discriminant Analysis (LDA) classifier. This model dynamically calibrates to the patient’s specific, reorganized neural patterns, effectively compensating for post-lesional cortical shifts and ensuring high decoding accuracy despite the altered brain activity.

4.3 Level of Autonomy, EEG Interface, and Inclusion Criteria

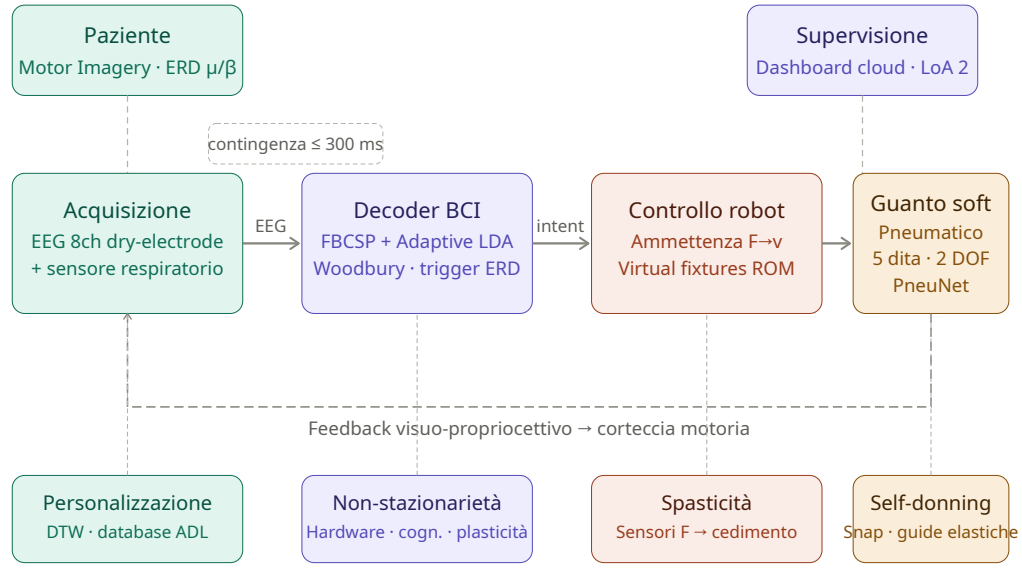
The system operates at **Level of Autonomy 2 (Shared Control)** according to the Troccaz classification: the patient retains high-level decisional control by generating motor intent, while the robot autonomously manages low-level execution (pressure trajectories, force sensor monitoring, and feedback timing).

The system cannot operate at LoA 3 or higher because autonomous actuation, decoupled from cortical intent, would violate the Hebbian plasticity principle. A robotic glove moving without the patient’s active intent would provide passive rehabilitation rather than neuroplastic training [4, 5]. Thus, LoA 2 is not a design compromise; it is the autonomy level required to induce neuroplasticity.

To ensure autonomous self-donning, the system utilizes a pre-shaped rigid-shell EEG headset with spring-loaded dry electrodes. The mechanical structure guides the headset into the correct position on the C3/C4 sites within 2 minutes, removing the need for conductive gel or high-precision alignment.

Inclusion Criteria as Design Specification Autonomous donning assumes sufficient function in the contralateral hand, which is implicit in the primary clinical inclusion criterion (patients must be capable of single-handed operation). The system is therefore indicated for patients with **unilateral hemiparesis**. Patients with severe bilateral paralysis are excluded by the device’s inherent design, not by a technological limitation. This clinical boundary is an intentional design specification.

Remote clinical supervision is conducted via an **asynchronous cloud-based dashboard**, allowing clinicians to regulate virtual fixtures (angular limits, admittance thresholds), review session logs, and personalize task parameters [12]. Patient progress is visualized through simplified traffic-light feedback cues displayed on a tablet.



Figur 2: Closed-loop architecture of the BCI-SRG system. Key components: (1) respiratory-synchronized EEG acquisition, (2) FBCSP-based adaptive LDA decoder, (3) safety controller integrating virtual fixtures and admittance control, and (4) the pneumatic soft glove. Dashed lines illustrate the visuo-proprioceptive feedback loop. Target end-to-end latency is ≤ 300 ms to satisfy Hebbian contingency requirements [4].

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